

VIJAYAKUMAR AATHIPAZHAM

Project Engineer — Costing, Estimation & EPC Execution | Oil & Gas | Petrochemical | Industrial Automation
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PROFESSIONAL SUMMARY

Results-driven Project Engineer with 10+ years of experience across EPC, Oil & Gas, Petrochemical, and industrial automation projects, with a strong focus on project costing, estimation, and techno-commercial evaluation. Proven track record preparing cost estimates, fabrication cost analyses, and RFQ/bid packages, alongside owning P&IDs/PFDs, engineering documentation, and multi-disciplinary project execution. Skilled in budget forecasting, supplier evaluation, variation order management, and cross-functional coordination between Finance, Engineering, Design, and Site teams to keep projects on schedule and within budget. Complements full-time EPC project engineering with independent freelance costing and estimation support engagements for structural, piping, static equipment, and instrumentation documentation scopes. Currently building toward leadership roles spanning project controls, engineering execution, and cross-border project delivery.

CORE COMPETENCIES & TECHNICAL SKILLS

- **Costing & Estimation:** Cost Estimation, Bill of Materials (BOM), Material Take-Off (MTO), Fabrication Cost Analysis, Budget Forecasting, Techno-Commercial Bid Evaluation, RFQ Management, Supplier/Vendor Evaluation, Variation Order & Change Cost Assessment, Piping Costing & Estimation, Structural Estimation
- **Engineering Tools:** SAP, COMOS cDB/iDB, AutoCAD, SolidWorks
- **Project Management Tools:** MS Project, Primavera P6 (Basic), MS Excel, Word, PowerPoint, ATG Programming
- **Core Competencies:** PFD & P&ID Ownership, Datasheet Development, Equipment & Material Selection, Engineering Documentation, ATEX Compliance, COMOS Training, Cross-Functional Team Collaboration, Instrumentation, Codes & Standards (API/ASME/ISO/ASTM)
- **Safety & Compliance Documentation:** Permit-to-Work (PTW) Documentation for Hazardous Area Work, Tank Testing, Critical Equipment Inspection, Maintenance Support, Instrument Testing
- **Equipment Expertise:** Chemical Equipment (Tanks, Valves, Pumps, Compressors, Heat Exchangers), Atmospheric & Chemical Storage Tanks, Static Equipment, Material Handling Systems & Instrumentation

PROFESSIONAL EXPERIENCE

Project Engineer — Mold Masters DME India Pvt Ltd (Coperion–GEC)

Aug 2025 – Present

- Own and execute dedicated project work packages end-to-end, coordinating with Project Managers and cross-disciplinary teams to meet schedule and quality targets.
- Contribute to project costing inputs and equipment/material selection trade-offs during FEED and EPC phases, aligning technical scope with budget and delivery expectations.
- Lead client and internal review meetings, driving timely resolution of bottlenecks and ensuring client feedback is addressed.
- Develop and maintain P&IDs and PFDs using COMOS (v10.2 & 10.4.1) and AutoCAD, ensuring full alignment with project specifications and EPC standards.
- Manage the complete PFD/P&ID lifecycle, integrating design changes and multi-discipline inputs in COMOS to deliver error-free documentation on time.
- Prepare detailed equipment datasheets meeting international specification standards, supporting accurate costing and procurement decisions.
- Support technical bid evaluations and deliver COMOS tool training to internal teams, standardizing practices and reducing dependency on external support.

Assistant Project Engineer — KJT

Nov 2021 – Mar 2025

- Led proposal preparation, costing, estimation, and fabrication cost analysis for engineering projects, translating technical scope into accurate, competitive project budgets.
- Prepared piping costing and estimation for atmospheric tanks and chemical storage equipment, covering material take-off, fabrication, and installation cost inputs for accurate bid pricing.
- Coordinated tank erection/testing and control panel erection/testing, including instrumentation installation, ensuring compliance with project specifications and quality standards.
- Prepared cable schedules and I/O lists for instrumentation and control systems, supporting accurate wiring, termination, and system integration.
- Conducted supplier evaluation, quotation comparison, and budget forecasting to support project cost control and margin protection throughout the project lifecycle.
- Reviewed contract POs to define project scope, objectives, and KPIs, aligning cost and execution plans with client requirements.

- Directed project engineering deliverables — drawings, datasheets, and BOMs — ensuring on-time submission, cost traceability, and full compliance with project schedules.
- Coordinated engineering deliverables across Finance, E&I, Design, and Engineering departments, and led technical clarifications with clients to secure approvals.
- Identified scope deviations and managed change requests, including techno-commercial variation orders and ECR/ECO approvals with cost and schedule impact assessment.
- Designed PFDs and created P&IDs detailing equipment, instrumentation, and piping layouts, diagnosing process issues to drive continuous optimization.
- Prepared safety documentation including permit-to-work (PTW) documents for hazardous area operations, supporting compliance with site safety protocols.
- Conducted tank testing, critical equipment inspection, and instrument testing, ensuring equipment readiness and adherence to quality and safety standards.
- Provided maintenance support across mechanical and instrumentation systems, coordinating corrective and preventive actions to minimize downtime.
- Maintained accurate design, costing, and technical modification records to support project documentation and audit readiness.

Junior Engineer (QMS & Project) — Autosys Engineering Pvt Ltd

Nov 2018 – Jul 2021

- Organized ISO 9001:2015 internal audits and led management review processes.
- Authored 30+ CAPA reports and led feedback analysis, resolving non-conformities and improving production line efficiency by 12%.
- Partnered with engineers to refine welding procedures, reducing defect rates by 18% within 6 months.
- Developed project plans, QAPs, detailed drawings, PFDs, and P&IDs from concept through design, contributing scope and material inputs used in project cost estimates.
- Provided site installation and testing support for major automation projects, including auto batching, conveying, dosing, and additive batching/weighing systems.

Quality Assurance Engineer — Sri Balaji Castings Pvt Ltd

Jun 2017 – Aug 2018

- Planned and executed internal quality audits, reporting findings to top management.
- Prepared meeting minutes and management reports on quality system performance.
- Conducted final inspections, dispatch clearance, calibration, and testing.

Trainee Engineer, Quality Assurance — TVS Sundaram Fasteners Pvt Ltd

Apr 2016 – Jun 2017

- Conducted inspection, dispatch clearance, and calibration activities.

KEY PROJECTS HANDLED

- **Raw Material Feeding – SPROB** | Client/Location: ISRO, Sriharikota | Scope: EPS | Role: Project Engineer
- **Additive Feeding & Bagging System – Brownfield** | Client/Location: Mahakoshal Refractories, Katni | Scope: EPC | Role: Project Engineer
- **Raw Material Handling & Bagging System – Greenfield** | Client/Location: RHI Magnesita (Vizag), IFGL (Vizag), Totale Global (Chennai) | Scope: EPC / Site Supervision | Role: Project Engineer
- **Extruder Packaging System** | Client/Location: BPCL–Kochi, GAIL PP, Alujain PP, Rongxin PP | Scope: FEED + EPC | Role: Project Engineer
- **Polyol Dosing System** | Client/Location: UNO Minda, Bawal | Scope: EPC | Role: Project Engineer (proposal costing & estimation)
- **Engineering Support & Maintenance** | Client/Location: Shell Petroleum, Chennai | Scope: EPC | Role: Project Engineer
- **Freelance Costing & Estimation Support** | Client/Location: Mefco Engineers Pvt Ltd, Ambattur; Autometric, Chennai | Scope: BOQ, MTO & Costing for Piping, Vessels, Material Handling & Static Equipment, Structural Estimation, Instrumentation Documentation | Role: Freelance Consultant (Nov 2021 – Apr 2025, Part-Time)

CERTIFICATIONS

- Internal Auditor, ISO 9001:2015
- Certified First Aider & Fire Fighter

EDUCATION

- B.E. Mechanical Engineering — Kings Engineering College, Chennai (2015), 75.41%
- 12th Grade — Karapettai Nadar Higher Secondary School, Tuticorin (2011), 84.41%
- 10th Grade — K.T. Kosalram Higher Secondary School, Tuticorin (2009), 88.6%